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## Bérince S. R. Hounsouvo

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Ph. D Candidate in Forest sciences, Specializing in Tree-Fungi-Sol Interactions | [Université Laval](#)

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### Education

#### Ph.D. in Forest Sciences

2023 - 2026

Department of wood and forest sciences, Université Laval, Québec Canada

*Thesis:* [Selection of boreal woody species with lateral root architecture for technosol-based revegetation of acid mine drainage sites](#)

*Supervisor:* [Prof. Damase P. Khasa](#)

*Co supervisor:* [Prof. Antoine Karam](#)

#### Master's degree, Forest Science and Techniques

2018 - 2020

Faculty of Agronomy, University of Parakou, (Benin)

*Thesis:* [Effect of ectomycorrhizal inoculation and organic and mineral amendments on juvenile growth of nursery plants of \*Azelia africana\*, \*Pterocarpus erinaceus\* & \*Khaya senegalensis\*.](#)

*Supervisor:* [Prof. Christine Ouinsavi](#)

#### Bachelor's degree, Natural Resource Management

2014 - 2017

Faculty of Agronomy, University of Parakou, (Benin)

*Thesis:* Optimal sampling technique for plot-based biodiversity measurement in the Sudanian zone: the case of square and circular plots

*Supervisor:* [Prof. Nourou S. Yorou](#)

### Additional Training

[5] [Big data processing and functional predictions in Biology - SymbioSol workshop 2025](#), Quebec, Canada.

[4] [NSERC CREATE Program on Nature-Based Solutions for Ecosystem Restoration - NASER Summer School 2025 \[in person\]](#). Université Laval, Quebec, Canada. From April 28 to May 09, 2025

[3] [Programming with R for data analysis \[in person\]](#). Université Laval, Québec, Canada.

[2] [Introduction to Unix Shell \[online, UNX101\]](#). Calcul Québec, Québec, Canada. February 2, 2024

[1] [Data analysis in bioinformatics: exploring computational biology \[in person\]](#). Université Laval, Quebec, Canada. From April 28 to May 09, 2025

### Research Interests

- Forest biology
- Mycorrhizal symbioses
- Forest ecophysiology
- Restoration ecology

### Academic and Research Experiences

#### Doctoral Researcher in forest biology and soil sciences

2023 - 2026

Department of wood and forest sciences, Université Laval, Québec Canada.

- Develop literature reviews on our scientific knowledge of tree-soil and tree-fungi interactions.
- Design and carry out advanced experimental protocols in forestry and restoration practices.
- Build statistical and mathematical R models to analyze biological data in forest sciences.

#### Teaching Assistant.

Winter 2025

Department of soils and agro-food engineering, Université Laval, Québec Canada

- Co-supervise a master's student in the design and implementation of a forest experiment
- Provide scientific guidance on experimental design and field data collection

#### **Research Assistant in Forest Biology**

2024 - 2026

Department of wood and forest sciences, Université Laval, Québec Canada.

- Assist a PhD candidate in designing a forest field experiment
- Participate in ECM (ectomycorrhizal) seedling inoculation

#### **Research Assistant in Forest Biology**

2021 - 2023

Forest Research and Studies Laboratory, University of Parakou, Parakou (Benin)

- Design and conduct forest experimentation projects
- Perform analysis of variance (ANOVA) and linear regression using R
- Write scientific and field mission reports

#### **Research Assistant in Tropical Mycology**

2017 - 2018

Research Unit in Tropical Mycology and Plant-Fungi-Soil Interactions, Parakou (Benin)

- Conduct surveys of macro-fungi in open forests and savannas
- Evaluate fungal diversity indices ( $\alpha$  and  $\beta$  diversity) using R software
- Manage fungi specimens in the Mycological Herbarium collection at the University of Parakou

### **Leadership Experiences**

#### **One Young World Ambassador**

2024 - 2025

One Young World, London Area, United Kingdom

- Exchange ideas, learn best practices, and collaborate with global leaders in conservation.

#### **Co-founder & President**

2021 - 2023

Environment Plan ONG, Benin, West Africa

- Definition of the vision and general policy of the NGO
- Elaboration and management of NGO's projects
- Management of work team and staff members

### **Publications**

[5] Hounsouvo, Bérince S. R., et al. Using mycorrhizal inocula for tree in Africa: an integrative conceptual framework and research roadmap (*in preparation for Biological reviews*).

[4] Hounsouvo, Bérince S. R., Khasa Damase P., Karam Antoine and Lamhamedi Mohammed S. Tree root development and architecture at post-mining sites: A comprehensive review (*in preparation for STOTEN*).

[3] Hounsouvo, Bérince S. R., Orou Sourou Zouliatou, Ayena Jacques I. K., and Dramani Ramdan. Socio-Agronomic Determinants of Farmland Access in Vegetable Production Systems in Northern Benin: Evidence from Bassila. Agriculture and Human Values (*submitted*). <https://doi.org/10.21203/rs.3.rs-6614725/v1>

[2] Hounsouvo, BSR., Houehanou, TD, Gouwakinnou, GN, Hadonou, CJ, and Ounsavi, CAIN. (2022). Comparative Effect of Ectomycorrhizal Inoculation, Mineral and Organic Amendments and their Interactions on Juvenile Growth of *Azadirachta indica* in the Nursery. Journal of Tropical Plant Physiology, 14(1), 1-12. 022):1-12 <https://doi.org/10.56999/jtpp.2022.14.1.18>

[1] Habakaramo M. P., Tchan K. I., Hegbe A. D. M. T., Abohombou G., Hounsouvo B. S. R., Tchemagnon O., Dramani R., and Yorou N. S. (2022). Sampling techniques for the optimal measurements of macromycetes diversity in the Soudano-Guinean ecozone (West Africa). International Journal of Biodiversity and Conservation, 14(3), 128-138. <https://doi.org/10.5897/IJBC2020.1404>

## Scientific Communications

- [4] *Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec*. Journée étudiante de Centre SÈVE, Quebec, Canada. [Poster]
- [3] *Variation génétique de la croissance racinaire chez les plants d'épinette noire et de mélèze laricin*. Colloque forestier 2025. L'office des producteurs de plants forestiers du Québec (OPPFQ) Quebec, Canada. [Oral presentation]
- [2] *Genetic Control and Repeatability of Root Growth in Seedlings of EPN and MEL, Two Species with Contrasting Strategies Along the Root Economics Spectrum*. Interlab 2025. Institute of Integrative Biology and Systems, Université Laval, Quebec, Canada. [Oral presentation]
- [1] *Root System Strategies of Picea mariana and Larix laricina for Revegetation of Acid Mine Drainage Sites in Quebec*. Journée étudiante de l'IBIS (15e ed), Institute of Integrative Biology and Systems, Université Laval, Quebec, Canada. [Poster]

## Awards - Grants - Fellowships (\$84,650)

### Doctoral degree - \$81,400

- [9] J. André Fortin grant in Forest Mycology and Forest Soil Biology: Influence of root morphology on the mycorrhizal community of boreal woody species (2026) - \$2500.
- [8] Doctoral thesis writing grant, Faculty of Graduate and Postdoctoral Studies, Université Laval (2026) - \$5000.
- [7] Best PhD progress award (37RAPF), Faculty of Graduate and Postdoctoral Studies, Université Laval (2025) - \$3700.
- [6] Participation Grant for the Student Day 2025, Centre SÈVE: Root System Strategies of *Picea mariana* and *Larix laricina* for Revegetation of Acid Mine Drainage Sites in Quebec (2025) - \$1000
- [5] Doctoral project completion Award (37EXAD), Faculty of Graduate and Postdoctoral Studies, Université Laval (2020) - \$6200
- [4] Award for doctoral project preparation, Faculty of Graduate and Postdoctoral Studies, Université Laval (2024) - \$3000
- [3] Doctoral fellowship, Department of wood and forest sciences, Université Laval (2023-2026) - \$60,000

### Master's degree - \$750

- [2] Master's fellowship, University of Parakou, Benin Republic (2019-2020) - \$750

### Bachelor's degree - \$2500

- [1] Bachelor's scholarship, Minister of Higher Education and scientific research, Benin Republic (2014-2017) - \$2500

## Mentoring

### Graduate co-advisor

- [3] Franel Affolabi 2025. Sélection des espèces à racines traçantes dans le contexte agroforestier post minier. Master's degree, Université Laval, Québec, Canada.
- [2] Justine Daviault 2025. *Pratiques de revégétalisation des sites miniers générateurs de drainage minier acide au québec : acquis, opportunités et défis*. Master's degree, Université Laval, Québec, Canada.

### Undergraduate co-advisor

- [1] Osé Zoundoh 2020. *Effets de différentes doses d'amendements organique et minérale sur la croissance juvénile de Afzelia, khaya et Pterocarpus*. Bachelor's degree, University of Parakou, Parakou, Benin